

Replay-Aware ATSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: atsp_failure_replay_compression.
- Best TSPLIB ATSP holdout gap: atsp_failure_replay_compression.
- Best synthetic holdout gap: atsp_random_replay.

Run Metadata

- run_name: run_20260513_174704_I
- started_at_local: 2026-05-13 17:47:04
- finished_at_local: 2026-05-13 17:58:46
- duration_hhmm: 00:12
- duration_seconds: 701.816
- seed_offset: 11000
- replicate_label: I
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB ATSP Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
atsp_no_replay	none	score_only	False	0.192622	0.009004	0.100813	0.889019	0.58	0.006947
atsp_random_replay	random	score_only	False	0.193105	0.008454	0.100779	0.832945	0.58	0.007474
atsp_failure_replay	failure	score_only	False	0.191028	0.009004	0.100016	0.843266	0.58	0.012535
atsp_failure_replay_compression	failure	novelty_gap	True	0.139721	0.020176	0.079949	0.731389	0.78	0.026014

Condition Notes

atsp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.192622, synthetic 0.009004, combined 0.100813.

- Accepted-epoch count 6, mean accepted code novelty 0.889019, and final complexity 0.58.
- Adaptation efficiency 0.006947 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.192622 across 4 instances; family means: ft=0.322375, ftv=0.33478, p=0.003915, ry=0.109416.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1903, "family": "ftv", "name": "ftv38", "optimality_gap": 0.243791}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1705, "family": "ftv", "name": "ftv35", "optimality_gap": 0.157502}]

atsp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.193105, synthetic 0.008454, combined 0.100779.
- Accepted-epoch count 6, mean accepted code novelty 0.832945, and final complexity 0.58.
- Adaptation efficiency 0.007474 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.193105 across 4 instances; family means: ft=0.323244, ftv=0.33478, p=0.004982, ry=0.109416.
- Panel synthetic_holdout mean gap 0.008454 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1903, "family": "ftv", "name": "ftv38", "optimality_gap": 0.243791}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1749, "family": "ftv", "name": "ftv35", "optimality_gap": 0.187373}]

atsp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.191028, synthetic 0.009004, combined 0.100016.
- Accepted-epoch count 4, mean accepted code novelty 0.843266, and final complexity 0.58.
- Adaptation efficiency 0.012535 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.191028 across 4 instances; family means: ft=0.323244, ftv=0.33478, p=0.004093, ry=0.101997.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1903, "family": "ftv", "name": "ftv38", "optimality_gap": 0.243791}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1749, "family": "ftv", "name": "ftv35", "optimality_gap": 0.187373}]

atsp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB ATSP 0.139721, synthetic 0.020176, combined 0.079949.
- Accepted-epoch count 3, mean accepted code novelty 0.731389, and final complexity 0.78.

- Adaptation efficiency 0.026014 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.139721 across 4 instances; family means: ft=0.222448, ftv=0.290763, p=0.000534, ry=0.045139.
- Panel synthetic_holdout mean gap 0.020176 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.080705, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1473, "cost": 1941, "family": "ftv", "name": "ftv35", "optimality_gap": 0.317719}, {"best_known_cost": 1530, "cost": 1966, "family": "ftv", "name": "ftv38", "optimality_gap": 0.284967}, {"best_known_cost": 1286, "cost": 1505, "family": "ftv", "name": "ftv33", "optimality_gap": 0.170295}]

Judge Appendix

Summary of Results for Replay-Aware ATSP Benchmark Suite

Condition	Synthetic Gap	TSP LIB Gap	Transfer Gap	Code Novelty (Mean)	Replay Mode	Complexity	Notes
No replay	0.009004	0.192622	0.100813	0.889	none	0.58	Highest TSPLIB gap, high code novelty, no replay
Random replay	0.008454	0.193105	0.100779	0.833	random	0.58	Similar transfer and TSPLIB gaps to no replay
Failure replay	0.009004	0.191028	0.100016	0.843	failure	0.58	Slightly improved transfer gap, similar TSPLIB gap
Failure replay + compression	0.020176	0.139721	0.079949	0.731	failure + compression	0.78	Best TSPLIB (0.1397) and transfer gaps (0.0799); lower code novelty; higher complexity

Interpretation

- ****Transfer evidence prioritization****:
- The ****failure replay + compression**** condition shows the ****best transfer performance**** indicated by the lowest mean held-out TSPLIB gap (0.1397) and transfer gap (0.0799).
- Synthetic holdout gap is worse in this condition (0.020176 vs ~0.009 in others), but the held-out realistic TSPLIB and holdout transfer gaps are more critical for transfer evaluation and show improvement.
- ****Replay Mode Effects****:
- Failure replay conditions (with or without compression) yield ****slightly better transfer gaps**** than no replay or random replay, which have nearly identical TSPLIB and transfer gaps.
- Compression pressure combined with failure replay further improves transfer results but at the cost of increased behavioral complexity (0.78 vs 0.58) and reduced code novelty (~0.73 vs ~0.83–0.89).
- ****Code Novelty vs Transfer****:

- Code novelty is highest for no replay (mean ~0.889) and random replay (~0.833), but these do not translate to improved transfer.
- Transfer improves in failure replay conditions despite a decrease in code novelty indicators.
- This indicates that the **improvements in transfer do not stem from increased code novelty**, suggesting that **lexical novelty is not aligned with algorithmic invention here**.
- **Complexity and Behavior**:
- The best performing transfer condition also corresponds to the highest complexity (0.78) and a "clustered_constructor" behavior profile.
- Other conditions maintain a consistent moderate complexity (0.58) and balanced behavior.

Conclusion

- **Failure replay with compression is the optimal replay-aware strategy to improve transfer performance on held-out TSPLIB ATSP benchmarks.**
- This approach reduces the mean transfer and TSPLIB optimality gaps significantly relative to no replay, random replay, and failure replay without compression.
- The strategy also incurs a moderate increase in complexity and a reduction in code novelty.
- Therefore, **improved transfer comes with diminished code novelty**, implying that code novelty is not a reliable proxy for algorithmic transfer performance in this setting.
- Random replay and no replay conditions exhibit similar transfer gaps and complexity but higher code novelty, indicating less efficient adaptation or informative replay conditioning.

Key Metrics (TSPLIB heldout panel mean optimality gaps):

Condition	TSPLIB Gap
No replay	0.1926
Random replay	0.1931
Failure replay	0.1910
Failure replay + compression	0.1397

Recommendations

- Prioritize **failure replay combined with compression pressure** for improved transfer in ATSP tasks despite increased complexity and lower novelty.
- Do not interpret code novelty metrics as evidence of algorithmic innovation without supportive transfer or optimality improvements.
- Distinguish replay modes carefully as failure replay variants yield better practical transfer even without lexical novelty gains.